FLEET TEAM DIGEST

737MAX-FTD-33-17001

Issue Title : Landing, Taxi and Runway Turn Off Lights (LTRTL) Degradation Overview

Airplane Model	ATA	Minor Model(s)
737MAX	3345-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
05/15/2017	09/04/2026	09/03/2026

Estimated Completion Date	Next Update Date

Revision Description

Added information on the redesigned LED LTRTL assemblies.

Description

The Landing, Taxi, and Runway Turnoff Light (LTRTL) is a new exterior lighting system introduced for 737NG production. The LTRTL system uses Light Emitting Diodes (LED) as the light source and replaces the existing halogen system. The system consolidates five installations into two, while increasing system reliability and reducing airline operating and maintenance costs. The LTRTL system for the 737MAX is unchanged and identical to the system used on the 737NG. This article provides details on how to manage the degradation of the LED lights as they age.

Background

General System Description:
• The LTRTL system is installed inside the left and right wing strakelets to provide the landing, taxi and runway turn off lighting functions. The following discussions are specific to the LTRTL components installed in one strakelet. The system components are mirror image in the other strakelet
• The 2 main LED arrays (inboard/outboard) provide the required lighting for landing and takeoff operations when operating at the max (100%) power setting.
• The same 2 main LED arrays provide the required lighting for taxi operations when operating at the lower (10%) power setting.
• The modes of operations for the main LED arrays are independently controlled by either selecting the Landing light switch or Taxi light switch to ON in the P5 overhead exterior lighting control panel.
• Since the same main LED arrays are being used to provide the required lighting for both taxi and landing operations, the landing light switch overrides the taxi light switch when both are selected ON.
• The Runway Turnoff (RTO) lighting function is provided via a separate RTO module.
• Power to the main LED arrays and RTO module is provided via a single power supply mounted directly behind the 2 main LED arrays.

Main LED Arrays Technical Information:
• Power to the main inboard and outboard LED arrays are provided by the LTRTL power supply via 2 separate/independent output channels.
• Each main LED array contains 36 LEDs. The original design has 3 separate vertical LED blades with each corresponding integrated reflectors that utilize 12 LEDs for a total of 36 LEDs. The redesigned main LED array assemblies uses Planar Arrays that consists of one LED Printed Board Assembly (PBA) that uses 36 LEDs.
• Electrical failures of a main LED array could result in individual LED(s) becoming in-op or the entire array (all 36 LEDs) becomes in-op.
• Each main array is designed so that there will be no significant impact to the overall lighting pattern or performance of the array

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with multiple random LEDs failures.

Power Supply Technical Information:
• There are 2 power supplies (PS) provided for the LTRTL system. Each power supply is installed in the left and right wing strakelet to provide independent power to the left/right main LED arrays and RTO.
• Each power supply contains 3 separate/independent output channels to provide the required electrical power to the inboard LED array, outboard LED array and the RTO module. Power is provided via 3 separate interconnecting wiring cables.
• Electrical failures of the power supply could affect 1 channel, 2 channels or all 3 channels at the same time.
• The power supply has an internal “fold back” circuitry to continuously monitor the main array LEDs temperature conditions and reduces power to the LED arrays to prevent potential LEDs degradation due to prolonged exposure to heat when operated under extreme ambient temperature conditions. The fold back condition is only predicted to occur at the system 100% max power setting (landing light mode) under extreme ambient temperatures. Analysis and qualification testing showed that the fold back condition only occurs under extreme ambient (surrounding air) temperature of 85C (185F), which is very unlikely for night or even daytime operations during the summer months.
• In the event of a loss of the temperature data from the main array(s), the power supply will default to a taxi light power setting of 10% (even in landing mode) to prevent continuous operations of the system at extreme temperature conditions that could potentially be damaging to the LEDs.

Runway Turn Off Light Technical Information:
• The original RTO assembly design contains 5 LEDs. The redesigned RTO assembly contains 20 LEDs.
• A Runway Turn Off Light may be considered to operate normally when the RTO assembly contains 1 or more inoperative LEDs.

Status

The LTRTL system was introduced in 737NG production starting Line Number 5555. The LTRTL system for the 737MAX is unchanged and identical to the system used on the 737NG.

Interim Action

None

Final Action

None

Milestones

Entry Into Service (EIS): September 2015

Part Information

LTRTL Degradation Information:

• Each main array can have up to 18 random LEDs failures with no maintenance action required. The built-in health monitoring circuit within the LTRTL power supply will automatically deactivate the array once the array voltage drops to a predetermined setting (50% of a nominal array voltage). If more than 18 LEDs are found to be non-operational, the main LED array is not serviceable in this condition and will need to be replaced.

• Qualification test and engineering analysis have been conducted to show that the intensity of the main LED array after its rated photometric life of 10,000 operational hours (approximately 24,000 flight hours) will still be at or above 80% of its initial intensity. In addition, a Boeing flight test was conducted with the system intensity at 80% and 50% and verified that the photometric performance of the system is acceptable for night operations even at 50% intensity.

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• There is no near end-of-life (EOL) or EOL indication provided in the LTRTL system to notify the pilots of the system's photometric degradation condition due to normal LEDs aging overtime. Additionally, unlike the navigation and anti-collision lights, there is no FAA requirements for the landing light to be at a certain intensity level. The only FAA requirement is that the lights have to provide enough light for night landing. Therefore, the airline pilots have the final authority to determine whether the aged/degraded main LED arrays (beyond 10,000 operational hours) can still provide adequate light for night operations.

• Unlike the main LED array, there is no health monitoring circuit to deactivate the RTO in the event of a loss of 1 or more LEDs. The LED failures within the RTO assembly is not serviceable and the entire RTO assembly has to be replaced. The airline pilots have the final authority to determine whether the RTO is still able to perform its intended function or needs to be replaced under these conditions.

Master Minimum Equipment List (MMEL) allowances:

• All are Repair Category C (Must be repaired within 10 consecutive calendar-days)

• Landing Lights - 3 options

4 Installed, 2 Required, One LED array (light assembly) may be inoperative on each side

4 Installed, 2 Required, Both LED arrays on the same side may be inoperative provided the Runway Turn Off light on the same side operates normally

4 Installed, 0 Required, May be inoperative for day operations.

• Taxi Lights: 4 Installed, 0 Required

• Runway Turnoff Lights: 2 Installed, 0 Required.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Component Maintenance Manual (CMM)	33-43-87 Power Supply		
Component Maintenance Manual (CMM)	33-43-88 RTO Assembly		
Component Maintenance Manual (CMM)	33-43-91 Main LED Array		
Aircraft Maintenance Manual (AMM)	33-42-21		

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability

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